Topic: Series solutions of higher order equations

Question: Find a power series solution in x to the differential equation.

$$y''' + y = 0$$

Answer choices:

A $$y = c_0 + c_1 x + c_2 x^2 + \frac{(-1)^{m+1} c_0}{(2m-2)!} x^{2m} + \frac{(-1)^m c_1}{(2m+4)!} x^{2m+1} + \frac{2(-1)^m c_2}{(2m+2)!} x^{2m+2} + \cdots$$

B $$y = c_0 + c_1 x + c_2 x^2 + \frac{(-1)^m c_0}{(3m+3)!} x^{3m+1} + \frac{(-1)^m c_1}{(3m+4)!} x^{3m+2} + \frac{3(-1)^m c_2}{(3m+5)!} x^{3m+3} + \cdots$$

C $$y = c_0 + c_1 x + c_2 x^2 + \frac{(-1)^m c_0}{(3m)!} x^{3m} + \frac{(-1)^m c_1}{(3m+1)!} x^{3m+1} + \frac{2(-1)^m c_2}{(3m+2)!} x^{3m+2} + \cdots$$

D $$y = c_0 + c_1 x + c_2 x^2 + \frac{(-1)^m c_0}{(4m)!} x^{4m} + \frac{(-1)^m c_1}{(4m+1)!} x^{4m+1} + \frac{3(-1)^m c_2}{(4m+2)!} x^{4m+2} + \cdots$$

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Solution: C

We'll substitute y and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=0}^{\infty} c_n x^n = 0$$

The series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, and $k = n$ into the second series.

$$\sum_{k=0}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + \sum_{k=0}^{\infty} c_k x^k = 0$$

With the series in phase and matching indices, we can now add them.

$$\sum_{k=0}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_k]x^k = 0$$

$$k = 0,\ 1,\ 2,\ 3,\ \ldots \qquad c_{k+3}(k+3)(k+2)(k+1) + c_k = 0$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3} = \frac{-c_k}{(k+3)(k+2)(k+1)}$$

Now we'll start plugging in values $k = 0,\ 1,\ 2,\ 3,\ \ldots$.

$$k = 0 \quad c_3 = \frac{-c_0}{(2)(3)} \qquad k = 1 \quad c_4 = \frac{-c_1}{(2)(3)(4)}$$

$$k = 2 \quad c_5 = \frac{-c_2}{(3)(4)(5)} \qquad k = 3 \quad c_6 = \frac{c_0}{(2)(3)(4)(5)(6)}$$

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$$k = 4 \quad c_7 = \frac{c_1}{(2)(3)(4)(5)(6)(7)} \qquad k = 5 \quad c_8 = \frac{c_2}{(3)(4)(5)(6)(7)(8)}$$

$$k = 6 \quad c_9 = \frac{c_0}{(2)(3)(4)(5)(6)(7)(8)(9)} \qquad k = 7 \quad c_{10} = \frac{-c_1}{(2)(3)(4)(5)(6)(7)(8)(9)(10)}$$

$$\vdots \qquad\qquad \vdots$$

We can start to see a pattern if we look at every third term. We can break the coefficients into three cases, given by c_{3m}, c_{3m+1}, and c_{3m+2} for $m = 1, 2, 3, \ldots$.

$$c_{3m} = \frac{(-1)^m c_0}{(3m)!} \qquad m = 1, 2, 3, \ldots$$

$$c_{3m+1} = \frac{(-1)^m c_1}{(3m+1)!} \qquad m = 1, 2, 3, \ldots$$

$$c_{3m+2} = \frac{2(-1)^m c_2}{(3m+2)!} \qquad m = 1, 2, 3, \ldots$$

Now we can write the general solution to the differential equation as

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + \ldots$$

$$y = c_0 + c_1 x + c_2 x^2 + c_{3m} x^{3m} + c_{3m+1} x^{3m+1} + c_{3m+2} x^{3m+2} + \cdots$$

$$y = c_0 + c_1 x + c_2 x^2 + \frac{(-1)^m c_0}{(3m)!} x^{3m} + \frac{(-1)^m c_1}{(3m+1)!} x^{3m+1} + \frac{2(-1)^m c_2}{(3m+2)!} x^{3m+2} + \cdots$$

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Topic: Series solutions of higher order equations

Question: Find a power series solution in x to the differential equation.

$$y''' + xy' + xy = 0$$

Answer choices:

A $$y = c_0\left(1 - \frac{1}{30}x^4 + \frac{3}{334}x^7 + \frac{1}{2{,}048}x^8 + \ldots\right)$$
$$+c_1\left(x - \frac{1}{24}x^3 - \frac{1}{60}x^4 + \frac{1}{1{,}260}x^6 + \frac{1}{2{,}688}x^7 + \frac{1}{30{,}240}x^8 + \ldots\right)$$
$$+c_2\left(x^2 - \frac{1}{15}x^5 - \frac{1}{10}x^6 + \frac{1}{1{,}008}x^8 + \frac{1}{10{,}080}x^9 + \ldots\right)$$

B $$y = c_0\left(1 - \frac{3}{8}x^5 + \frac{4}{7}x^8 + \frac{5}{6}x^{10} + \ldots\right)$$
$$+c_1\left(x - \frac{1}{3}x^4 - \frac{2}{3}x^5 + \frac{4}{3}x^6 + \frac{10}{7}x^7 + \frac{1}{3}x^9 + \ldots\right)$$
$$+c_2\left(x^2 - \frac{2}{11}x^7 - \frac{4}{9}x^8 + \frac{4}{13}x^{10} + \frac{5}{16}x^{11} + \ldots\right)$$

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C $$y = c_0\left(1 - \frac{1}{12}x^4 + \frac{1}{630}x^7 + \frac{1}{4{,}032}x^8 + \ldots\right)$$

$$+c_1\left(x - \frac{1}{48}x^4 - \frac{1}{120}x^5 + \frac{1}{2{,}320}x^7 + \frac{1}{5{,}376}x^8 + \frac{1}{60{,}480}x^9 + \ldots\right)$$

$$+c_2\left(x^2 - \frac{1}{30}x^5 - \frac{1}{120}x^6 + \frac{1}{2{,}016}x^8 + \frac{1}{6{,}048}x^9 + \ldots\right)$$

D $$y = c_0\left(1 - \frac{1}{24}x^4 + \frac{1}{1{,}260}x^7 + \frac{1}{8{,}064}x^8 + \ldots\right)$$

$$+c_1\left(x - \frac{1}{24}x^4 - \frac{1}{60}x^5 + \frac{1}{1{,}260}x^7 + \frac{1}{2{,}688}x^8 + \frac{1}{30{,}240}x^9 + \ldots\right)$$

$$+c_2\left(x^2 - \frac{1}{30}x^5 - \frac{1}{120}x^6 + \frac{1}{2{,}016}x^8 + \frac{1}{6{,}048}x^9 + \ldots\right)$$

Solution: D

We'll substitute y, y', and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + x\sum_{n=1}^{\infty} c_n n x^{n-1} + x\sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

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The first series begins with the x^0 term while the second and third series both begin with the x^1 term. So to put these series in phase, we'll pull the x^0 term out of the first series.

$$c_3 3(3-1)(3-2)x^{3-3} + \sum_{n=4}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^n + \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, $k = n$ into the second series, and $k = n + 1$ and $n = k - 1$ into the third series.

$$6c_3 + \sum_{k=1}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + \sum_{k=1}^{\infty} c_k k x^k + \sum_{k=1}^{\infty} c_{k-1} x^k = 0$$

With the series in phase and matching indices, we can now add them.

$$6c_3 + \sum_{k=1}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_k k + c_{k-1}]x^k = 0$$

The $6c_3$ value in front of the series is associated with the $k = 0$ term, while the remaining series still represents the $k = 1, 2, 3, \ldots$ terms.

$$k = 0 \qquad 6c_3 = 0$$

$$c_3 = 0$$

$$k = 1, 2, 3, \ldots \qquad c_{k+3}(k+3)(k+2)(k+1) + c_k k + c_{k-1} = 0$$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

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$$c_{k+3} = \frac{-c_{k-1} - c_k k}{(k+3)(k+2)(k+1)}$$

Now we'll start plugging in values $k = 1, 2, 3, \ldots$.

$$k = 0 \qquad c_3 = 0$$

$$k = 1 \qquad c_4 = \frac{-c_0 - c_1}{24} \qquad k = 2 \qquad c_5 = \frac{-c_1 - 2c_2}{60}$$

$$k = 3 \qquad c_6 = \frac{-c_2}{120} \qquad k = 4 \qquad c_7 = \frac{c_0 + c_1}{1{,}260}$$

$$k = 5 \qquad c_8 = \frac{c_0 + 3c_1 + 4c_2}{8{,}064} \qquad k = 6 \qquad c_9 = \frac{c_1 + 5c_2}{30{,}240}$$

$$\vdots \qquad\qquad \vdots$$

There's not an obvious pattern here. As we've seen, the higher the order of the differential equation, the less likely it becomes that we'll be able to find clean series representations.

So without worrying about the pattern, we'll just use these values to say that the general solution is

$$y = c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 + c_6 x^6 + \ldots$$

$$y = c_0 + c_1 x + c_2 x^2 + \frac{-c_0 - c_1}{24} x^4 + \frac{-c_1 - 2c_2}{60} x^5 + \frac{-c_2}{120} x^6 + \ldots$$

$$y = c_0 \left(1 - \frac{1}{24} x^4 + \frac{1}{1{,}260} x^7 + \frac{1}{8{,}064} x^8 + \ldots \right)$$

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$$+c_1\left(x-\frac{1}{24}x^4-\frac{1}{60}x^5+\frac{1}{1{,}260}x^7+\frac{1}{2{,}688}x^8+\frac{1}{30{,}240}x^9+\ldots\right)$$

$$+c_2\left(x^2-\frac{1}{30}x^5-\frac{1}{120}x^6+\frac{1}{2{,}016}x^8+\frac{1}{6{,}048}x^9+\ldots\right)$$

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Topic: Series solutions of higher order equations

Question: Find a power series solution in x to the differential equation.

$$y''' + x^2y' + xy = 0$$

Answer choices:

A $$y = c_0 + c_1x + c_2x^2 + \frac{(-1)^m 1^2 5^2 \cdots (2m-3)^2 c_0}{(2m)!}x^{2m}$$

$$+\frac{(-1)^m 2^2 6^2 \cdots (2m-2)^2 c_1}{(2m+1)!}x^{2m+1}$$

$$+\frac{2(-1)^m 3^2 7^2 \cdots (2m-1)^2 c_2}{(2m+2)!}x^{2m+2} + \cdots$$

B $$y = c_0 + c_1x + c_2x^2 + \frac{(-1)^m 2^2 8^2 \cdots (4m-3)^2 c_0}{(4m-2)!}x^{4m}$$

$$+\frac{(-1)^m 3^2 9^2 \cdots (4m-4)^2 c_1}{(4m-3)!}x^{4m+1}$$

$$+\frac{2(-1)^m 4^2 11^2 \cdots (4m-5)^2 c_2}{(4m-6)!}x^{4m+2} + \cdots$$

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C $$y = c_0 + c_1x + c_2x^2 + \frac{(-1)^m 2^2 5^2 \cdots (3m-2)^2 c_0}{(3m)!}x^{3m}$$

$$+\frac{(-1)^m 3^2 6^2 \cdots (3m-2)^2 c_1}{(3m+1)!}x^{3m+1}$$

$$+\frac{2(-1)^m 4^2 7^2 \cdots (3m-1)^2 c_2}{(3m+2)!}x^{3m+2} + \cdots$$

D $$y = c_0 + c_1x + c_2x^2 + \frac{(-1)^m 1^2 5^2 \cdots (4m-3)^2 c_0}{(4m)!}x^{4m}$$

$$+\frac{(-1)^m 2^2 6^2 \cdots (4m-2)^2 c_1}{(4m+1)!}x^{4m+1}$$

$$+\frac{2(-1)^m 3^2 7^2 \cdots (4m-1)^2 c_2}{(4m+2)!}x^{4m+2} + \cdots$$

Solution: D

We'll substitute y, y', and y''' into the differential equation.

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + x^2 \sum_{n=1}^{\infty} c_n n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0$$

$$\sum_{n=3}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^{n+1} + \sum_{n=0}^{\infty} c_n x^{n+1} = 0$$

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The first series begins with the x^0 term, the second series begins with the x^2 term, and the third series begins with the x^1 term. So to put these series in phase, we'll pull the x^0 and x^1 terms out of the first series and pull the x^1 term out of the third series.

$$c_3 3(3-1)(3-2)x^{3-3} + c_4 4(4-1)(4-2)x^{4-3} + \sum_{n=5}^{\infty} c_n n(n-1)(n-2)x^{n-3}$$

$$+ \sum_{n=1}^{\infty} c_n n x^{n+1} + c_0 x^{0+1} + \sum_{n=1}^{\infty} c_n x^{n+1} = 0$$

$$6c_3 + (c_0 + 24c_4)x + \sum_{n=5}^{\infty} c_n n(n-1)(n-2)x^{n-3} + \sum_{n=1}^{\infty} c_n n x^{n+1} + \sum_{n=1}^{\infty} c_n x^{n+1} = 0$$

Now the series are in phase, but the indices don't match. We can substitute $k = n - 3$ and $n = k + 3$ into the first series, and $k = n + 1$ and $n = k - 1$ into the second and third series.

$$6c_3 + (c_0 + 24c_4)x + \sum_{k=2}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k$$

$$+ \sum_{k=2}^{\infty} c_{k-1}(k-1)x^k + \sum_{k=2}^{\infty} c_{k-1}x^k = 0$$

With the series in phase and matching indices, we can finally add them.

$$6c_3 + (c_0 + 24c_4)x + \sum_{k=2}^{\infty} c_{k+3}(k+3)(k+2)(k+1)x^k + c_{k-1}(k-1)x^k + c_{k-1}x^k = 0$$

$$6c_3 + (c_0 + 24c_4)x + \sum_{k=2}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_{k-1}(k-1) + c_{k-1}]x^k = 0$$

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$$6c_3 + (c_0 + 24c_4)x + \sum_{k=2}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_{k-1}(k-1+1)]x^k = 0$$

$$6c_3 + (c_0 + 24c_4)x + \sum_{k=2}^{\infty} [c_{k+3}(k+3)(k+2)(k+1) + c_{k-1}k]x^k = 0$$

This equation gives us

$k = 0$ $\quad 6c_3 = 0$

$c_3 = 0$

$k = 1$ $\quad c_0 + 24c_4 = 0$

$$c_4 = \frac{-c_0}{(2)(3)(4)}$$

$k = 2, 3, 4, \ldots$ $\quad c_{k+3}(k+3)(k+2)(k+1) + c_{k-1}k = 0$

We'll solve the recurrence relation for the coefficient with the largest subscript, c_{k+3}.

$$c_{k+3} = -\frac{c_{k-1}k}{(k+3)(k+2)(k+1)}$$

Now we'll start plugging in values $k = 2, 3, 4, \ldots$. In this step, it's best to keep the multiplications written out rather than calculating them so we can try to find identify a pattern.

$k = 0 \quad c_3 = 0$

$k = 1 \quad c_4 = \dfrac{-c_0}{(2)(3)(4)}$

$k = 2 \quad c_5 = \dfrac{-2c_1}{(3)(4)(5)}$

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$$k = 3 \quad c_6 = \frac{-3c_2}{(4)(5)(6)} \qquad k = 4 \quad c_7 = 0$$

$$k = 5 \quad c_8 = \frac{5c_0}{(2)(3)(4)(6)(7)(8)} \qquad k = 6 \quad c_9 = \frac{(2)(6)c_1}{(3)(4)(5)(7)(8)(9)}$$

$$k = 7 \quad c_{10} = \frac{(3)(7)c_2}{(4)(5)(6)(8)(9)(10)} \qquad k = 8 \quad c_{11} = 0$$

$$k = 9 \quad c_{12} = \frac{-(5)(9)c_0}{(2)(3)(4)(6)(7)(8)(10)(11)(12)}$$

$$k = 10 \quad c_{13} = \frac{-(2)(6)(10)c_1}{(3)(4)(5)(7)(8)(9)(11)(12)(13)}$$

$$k = 11 \quad c_{14} = \frac{-(3)(7)(11)c_2}{(4)(5)(6)(8)(9)(10)(12)(13)(14)}$$

$$\vdots$$

We can start to see a pattern if we look at every fourth term. We can break the coefficients into four cases, given by c_{4m}, c_{4m+1}, c_{4m+2}, and c_{4m+3} for $m = 1, 2, 3, \ldots$.

$$c_{4m} = \frac{(-1)^m 1^2 5^2 \cdots (4m-3)^2 c_0}{(4m)!} \qquad m = 1, 2, 3, \ldots$$

$$c_{4m+1} = \frac{(-1)^m 2^2 6^2 \cdots (4m-2)^2 c_1}{(4m+1)!} \qquad m = 1, 2, 3, \ldots$$

$$c_{4m+2} = \frac{2(-1)^m 3^2 7^2 \cdots (4m-1)^2 c_2}{(4m+2)!} \qquad m = 1, 2, 3, \ldots$$

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$$c_{4m+3} = 0$$

Now we can write the general solution to the differential equation as

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + c_6x^6 + \dots$$

$$y = c_0 + c_1x + c_2x^2 + c_3x^3 + c_{4m}x^{4m} + c_{4m+1}x^{4m+1} + c_{4m+2}x^{4m+2} + c_{4m+3}x^{4m+3} + \cdots$$

$$y = c_0 + c_1x + c_2x^2 + \frac{(-1)^m 1^2 5^2 \cdots (4m-3)^2 c_0}{(4m)!}x^{4m}$$

$$+ \frac{(-1)^m 2^2 6^2 \cdots (4m-2)^2 c_1}{(4m+1)!}x^{4m+1}$$

$$+ \frac{2(-1)^m 3^2 7^2 \cdots (4m-1)^2 c_2}{(4m+2)!}x^{4m+2} + \cdots$$

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